

VIJAYAVARMAN ARUNASALAM HARIKRISHNAN

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EDUCATION

Master of Science in Cybersecurity

Rochester Institute of Technology, Rochester, NY

GPA – 3.97/4.0

Aug 2023-May 2026

Bachelor of Engineering in Computer Science and Engineering

Meenakshi College of Engineering, Chennai, India

GPA – 8.3/10.0

Aug 2016-Apr 2020

PROFESSIONAL EXPERIENCE

Rochester Regional Health, Rochester

Identity and Access Management Intern

Aug 2025-Present

- Process IAM service requests through Ivanti including verifying and executing password resets via Active Directory after confirming legitimacy with users and their managers
- Onboard and manage user, service, and vendor accounts in CyberArk PAM, configuring safes and permissions aligned with least-privilege access and supporting SailPoint identity governance implementation
- Develop PowerShell automation to monitor AD password reset events and notify affected users and maintain a C# IAM portal using Entity Framework and Azure DevOps

Cognizant Technology Solutions, Chennai

Programmer Analyst Trainee / Analyst / Associate

Nov 2020-Aug 2023

- Automated API and functional testing workflows using Selenium, improving efficiency and application reliability
- Developed and executed 50+ test cases per release, resolving critical defects with cross-team collaboration
- Mentored 5 new team members on testing workflows and tools to accelerate their onboarding and productivity

TEACHING EXPERIENCE

Rochester Institute of Technology

Teaching Assistant – Cryptography and Auth Aug 2024-Dec 2024, Aug 2025-Dec 2025

- Managed, graded, and provided feedback on Python and Java programming assignments covering cryptographic algorithms (Substitution Cipher, RC4, A5/1, MD5), using GitHub Classroom for submission and evaluation

Rochester Institute of Technology

Teaching Assistant – Introduction to Cybersecurity

Jan 2025-Aug 2025

- Assisted 60+ students with lab sessions and assignments covering network security, risk analysis, access control, and system security and graded 25+ assignments per term with detailed feedback

PUBLICATION

S. M. Maheen, I. Sultana, M. N. Fardous Zim, V. Arunasalam Harikrishnan and N. Kshetri, "ARTblock: Blockchain-Integrated AI for Real-Time Transaction Authenticity Verification in FinTech," *2025 International Conference on Electrical and Computer Engineering Researches (ICECER)*, Antananarivo, Madagascar, 2025, pp. 1-6, doi: 10.1109/ICECER65523.2025.11400953

PROJECTS

PQC-DEMS: Post-Quantum Cryptographic Digital Evidence Management System

Apr 2026

- Built a functional web-based forensic evidence management system implementing NIST-standardized ML-KEM-768 and ML-DSA-65 with RBAC, JWT auth, and a hash-chained audit log, benchmarked across file sizes up to 1GB
- Compared PQC and RSA across all operations confirming PQC superiority in key generation, signing, upload, and download while documenting RSA verification as 4 to 5x faster, a known lattice tradeoff

Forensic Recovery of Web Browser Artifacts

Apr 2025

- Investigated browser artifact recovery across Chrome, Firefox, and Edge under normal, deleted, and private browsing conditions using FTK Imager, Autopsy, and Volatility to analyze digital evidence persistence

IDOTS+: Enhancing IoT Dots for Smart Environments

May 2024

- Built an IoT forensics framework using a CNN log analyzer on OpenHAB and Raspberry Pi achieving 89% accuracy in detecting malicious activity in smart home environments

Attack and Defend Exercise – Securing Corporate Infrastructure

Apr 2024

- Deployed a multi-service corporate network monitored with Wazuh and Snort analyzing 26,000+ security events and conducted controlled attacks including EternalBlue, SQL Injection, and Path Traversal

CERTIFICATIONS

- CompTIA Security+ (SY0-701)
- Fortinet Certified Fundamentals in Cybersecurity
- Google Cybersecurity Professional Certificate
- Fortinet Certified Associate in Cybersecurity

COMPETITIONS

- NCAE Cyber Games 2025 – Finished 3rd among the top 12 teams from the East Overflow region
- CyberSeed 2024 CTF – Finished 23rd among 160+ teams after completing 87% of challenges

SKILLS

- Languages and Scripting:** Python, PowerShell, C, Java, SQL
- Platforms and Systems:** Active Directory, Azure AD (Entra ID), CyberArk PAM, SailPoint, Azure DevOps
- Security Tools:** Wazuh, Snort, Wireshark, Nmap, Burp Suite, Metasploit, FTK Imager, Autopsy, Volatility, KQL
- Frameworks and Standards:** NIST CSF 2.0, NIST RMF, HIPAA, PCI DSS, GDPR
- Cryptography:** ML-KEM-768, ML-DSA-65, AES-256-GCM, ElGamal
- Development:** FastAPI, SQLite, GitHub Classroom, Selenium, MATLAB